Expt: 9
Date: 11/9/26

Implementation of Subnetting in Cisco Packet Tracer.

Aim:

To design multiple subnets with suitable host capacities, assign static IP address, connect the subnets through a router, and simulate packet transmission with and without a router.

Algorithm / procedure:

1. Open Cisco Packet Tracer
2. Place PCs, Switches, and one router in the workspace.
3. Divide the network into two or more subnets based on the required number of hosts
4. Assign each subnet a unique network address and subnet mask.
5. Assign static IP addresses, subnet masks, and default gateway addresses to all PCs.
6. Configure the router interface with IP addresses belonging to the respective subnets.
7. Connect the PCs to switches and connect each switch to the router using suitable ethernet cables.

Command Prompt

```
PC>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=16ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=11ms TTL=128
Reply from 192.168.1.11: bytes=32 time=6ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 16ms, Average = 8ms

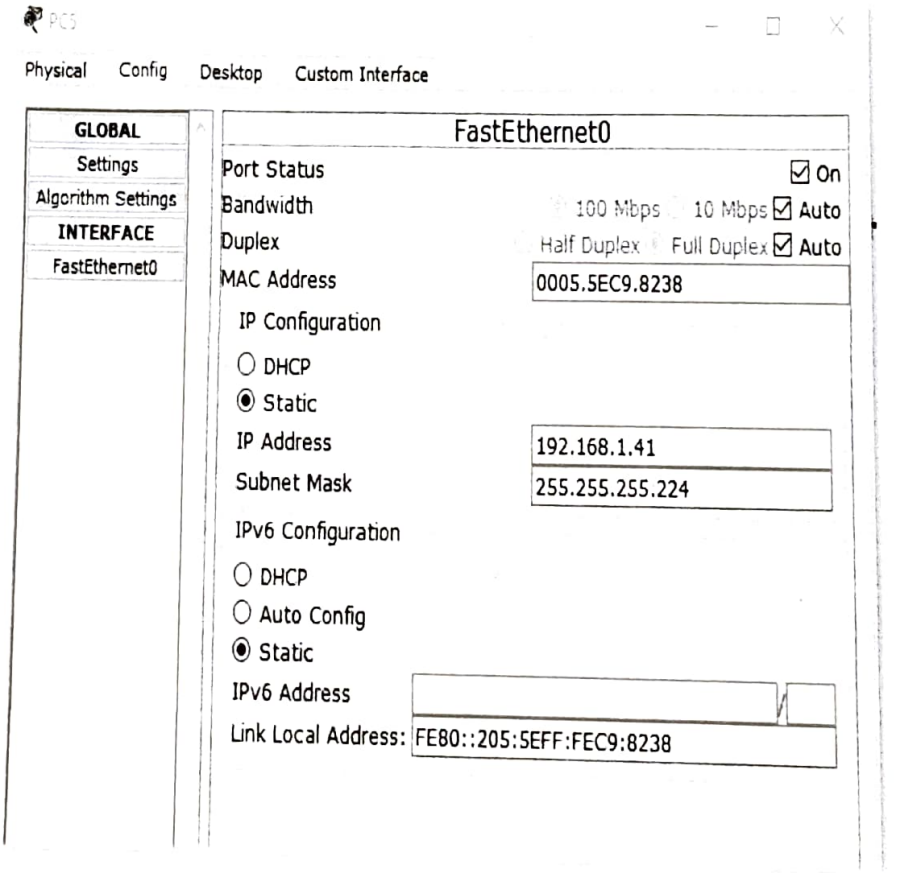
PC>ping 192.168.1.12

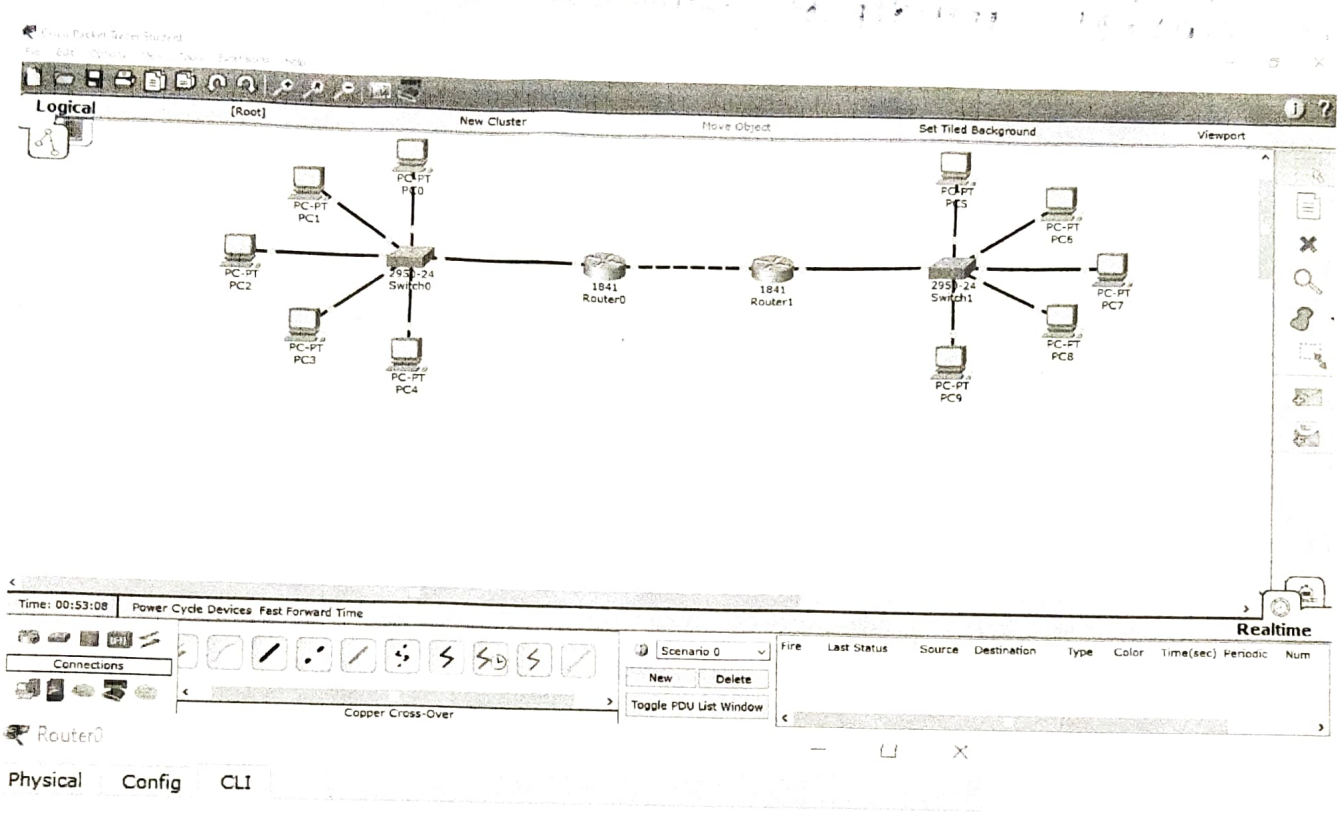
Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time=41ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=0ms TTL=128
Reply from 192.168.1.12: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 41ms, Average = 10ms

PC>
```





IOS Command Line Interface

```
exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.1.1 255.255.255.224
% 192.168.1.0 overlaps with FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.1.65 255.255.255.224
% 192.168.1.64 overlaps with FastEthernet0/1
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#do show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol

FastEthernet0/0        192.168.1.1     YES manual up                    down

FastEthernet0/1        192.168.1.65    YES manual up                    up

Vlan1                  unassigned      YES unset  administratively down down
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to
up
```

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8. Enable the router interfaces using the no shutdown command.

9. Use the Simulation mode or the ping command to test communication b/w PCs in different subnets.

10. Disconnect or disable the router connections and repeat the ping test.

11. Compare the results in both cases.

Result:

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Multiple subnets were designed and configured successfully. Packet transmission b/w different subnets was successful when the router was configured correctly. Communication between the subnets failed when the router connection was removed and no alternative routing path was available.